Sets and Functions

- **Sets**
- Functions
- Cardinality of Sets

- Reading: Rosen 2.1, 2.2, 2.3, 2.5

Set

- **Definition:** A **set** is an **unordered** collection of objects The objects in a set are called the **elements** or **members** of the set. A set is said to contain its elements. We write a $a \in A$ to denote that a is an element of the set A and $a \notin A$ to denote that a is not an elements of A.
- **Roster method for describing sets:**
  - List all elements of a set explicitly
  - Use … when the pattern is obvious.
- **Example**
  - The set of all odd positive integers less than 10 can be denoted by $\{1, 3, 5, 7, 9\}$.
  - The set of all odd positive integers less than 100 can be denoted by $\{1, 3, 5, \ldots, 99\}$.

Some Important Sets

$\mathbf{N}$ = *natural numbers* = $\{0, 1, 2, 3, \ldots\}$

$\mathbf{Z}$ = *integers* = $\{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\}$

$\mathbf{Z}^+$ = *positive integers* = $\{1, 2, 3, \ldots\}$

$\mathbf{Z}^-$ = *negative integers* = {-1, -2,- 3, ...}

$\mathbf{R}$ = *real numbers*

$\mathbf{R}^+$= *positive real numbers*

$\mathbf{R}^-$= *negative real numbers*

$\mathbf{Q}$ = *rational numbers*

Set Builder Method

- We can also use the **set builder** notation to express the set as $\{x \mid x \text{ is an odd positive interger less than } 100\}$, or $\{x \mid x \in \mathbf{Z}^+ \text{ and x is odd and } x < 100\}$,
- Generally, we can define a set as $\{x \mid P(x)\}$ or $\{x : P(x)\}$ using a predicate $P(x)$, which contains all x such that $P(x) = T$.
- Remark: Using set builders may lead to **paradoxes**
- A village has a single male barber who follows a strict rule: he shaves all men who **do not shave themselves**, and **only** those men.
- **Russel's paradox:** Who shaves the barber?
- **If the barber shaves himself:** Then he breaks rule, because he is only supposed to shave men who do not shave themselves.
- **If the barber does not shave himself:** Then he must be shaved by the barber (i.e., by himself), breaking the rule that he shaves everyone who doesn't shave themselves.
- Leads to axiomatic set theory. In this course we study "*naïve set theory*"

Empty Set and Singleton Set

- **Definition**
  The **empty set** or **null set**, denoted by ∅ or {}, is a special set containing no elements.
- **Definition**
  A set with one element is called a **singleton set**.
- **Example**
  Note that $\emptyset \neq \{\emptyset\}$. The latter is a singleton set.
  Yes, an element of a set can also be a set!
- In pure set theory, everything is a set!

Set Equality

- **Definition**

  Two sets A and B are **equal**, denoted by $A = B$, if and only if they have the same elements, i.e., for every x, $x \in A$ if and only if $x \in B$.

- **Example 3**

$$\{1, 2, 3\} = \{3, 1, 2\} = \{1, 2, 2, 2, 3, 3\}$$

Subset

- **Definition**

  A set A is said to be a **subset** of a set B, denoted by $A \subseteq B$, if and only if every element of A is also an element of B.

- **Note**

  For every set S, (a) $\emptyset \subseteq S$ and (b) $S \subseteq S$.

  Every nonempty set is guaranteed to have at least two subsets, the empty set and the set itself.

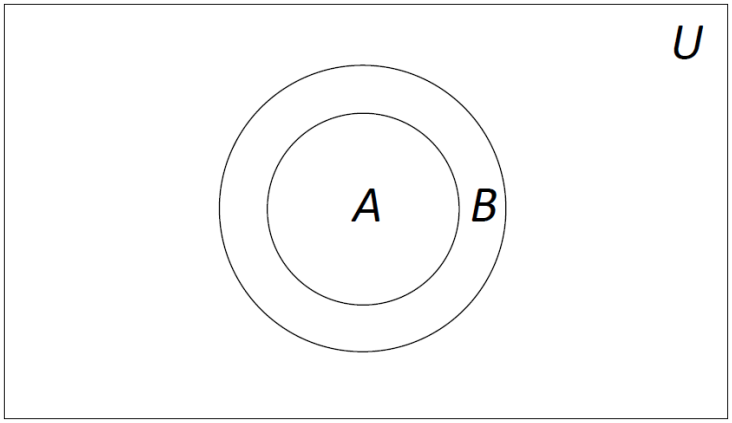
- $A = B$ if and only if $A \subseteq B$ and $B \subseteq A$.

Proper Subset

- **Definition**

  A set A is said to be a **proper subset** of a set B, denoted by $A \subset B$, if and only if $A \subseteq B$ but $A \neq B$.

We can use a **Venn diagram** to illustrate, among other things, the subset relationship.



Cardinality of Finite Sets

- **Definition**
  Let S be a finite set. The cardinality of S, denoted by $|S|$, is the number of (distinct) elements in S.
- **Definition**
  A set is said to be infinite if it is not finite.

Power Set

- **Definition**
  Given a set S, the power set of S, denoted by $P(S)$, is the set of all subsets of S.
- **Example**
  The power set of the set $\{a, b, c\}$ is $P(\{a, b, c\}) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$.
- **Remark**
  If a set has n elements where n is a nonnegative integer, then its power set has 2^n elements.
- **Example**
- What is the power set of the empty set?
- $P(\emptyset) = \{\emptyset\}$ (cardinality $1 = 2^0$)
- What is the power set of the set $\{a\}$?
- $P(\{a\}) = \{\emptyset, \{a\}\}$ (cardinality $2 = 2^1$)

Ordered Tuple

- **Definition**

  The **ordered *n*-tuple** $(a_1, a_2, \dots, a_n)$ is the ordered collection that has a_1 as its first element, a_2 as its second element, …, and $\boldsymbol{a_n}$ as its n-th elements. An ordered 2-tuple is more commonly called an **ordered pair**. Informally, an ordered tuple can be called and ordered set.

- **Definition**

  Two ordered n-tuples $(a_1, a_2, \dots, a_m)$ and $(b_1, b_2, \dots, b_n)$ are equal if and only if $m = n$ and $a_i = b_i$ for all $i = 1,2, \dots, n$.

Cartesian Product of Two Sets

- **Definition**

  Let A and B be two sets. The **Cartesian product** of A and B, denoted by $A \times B$, is the set of all ordered pairs (a, b) where $a \in A$ and $b \in B$, i.e.

  $$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}.$$

- **Definition**

  Let A and B be two sets. A subset R of $A \times B$ is called a **relation** from the set A to the set B.

Cartesian Product Example

- **Example**

  Let $A = \{1,2\}$ and $B = \{a, b, c\}$. The Cartesian product is

  $$A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}$$

  and

  $R = \{(1, a), (1, c), (2, a), (2, b)\} \subset A \times B$ is an example relation from A to B.

Cartesian Product of Multiple Sets

- **Definition**

  The **Cartesian product** of the sets $A_1, A_2, \ldots, A_n$ denoted by $A_1 \times A_2 \times \cdots \times A_n$, is the set of ordered *n*-tuples $(a_1, a_2, \ldots, a_n)$, where a_i belongs to A_i for $i = 1,2, \ldots, n$, i.e.

$$A_1 \times A_2 \times \cdots \times A_n = \{(a_1, a_2, \ldots, a_n) \mid a_i \in A_i \text{ for } i = 1,2, \ldots, n\}.$$

Union

- **Definition**

  Let A and B be two sets. The **union** of A and B, denoted by $A \cup B$, is the set that contains those elements that are either in A or in B, or in both.

- $A \cup B = \{x \mid x \in A \vee x \in B\}$

- **Definition**

  Let $A_1, A_2, \ldots, A_n$ be *n* sets. The **union** of the collection of *n* sets, denoted by $\cup_{i=1}^{n} A_i$, is the set that contains those elements that are members of at least one set in the collection.

Intersection

- **Definition**

  Let A and B be two sets. The **intersection** of A and B, denoted by $A \cap B$, is the set that contains those elements that are in both A and B.

- $A \cap B = \{x \mid x \in A \wedge x \in B\}$

- **Definition**

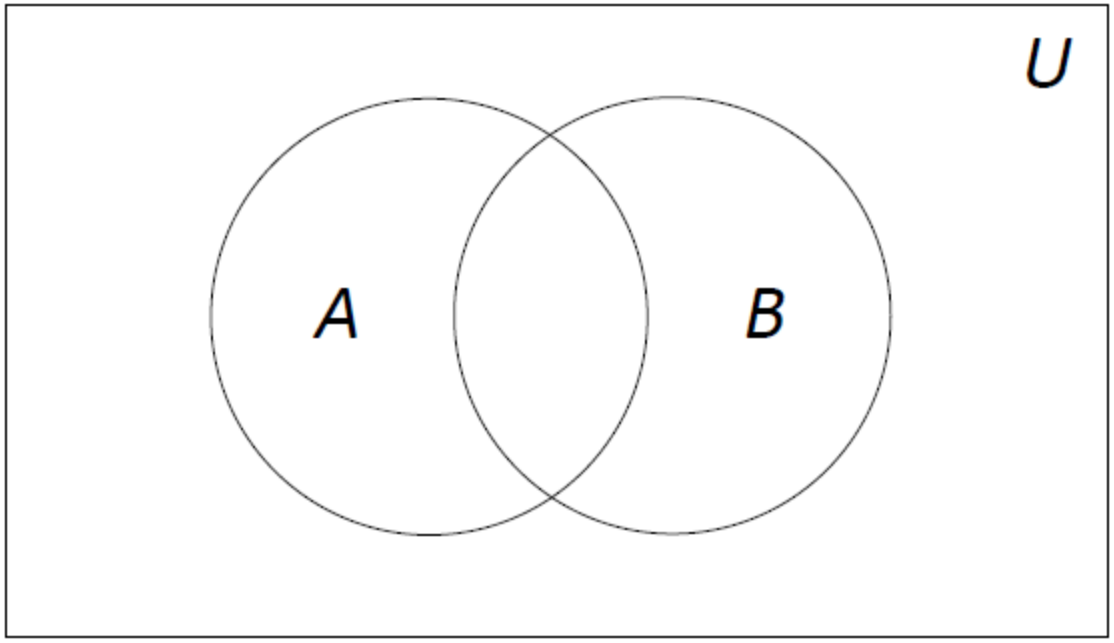
  Let $A_1, A_2, \ldots, A_n$ be *n* sets. The **intersection** of the collection of *n* sets, denoted by $\cap_{i=1}^{n} A_i$, is the set that contains those elements that are members of all the sets in the collection.

- **Definition**

  Two sets A and B are **disjoint** if $A \cap B = \emptyset$.

Union, Intersection, and Cardinality

- **Theorem**
  Let A and B be two finite sets. The cardinality of their union $|A \cup B| = |A| + |B| - |A \cap B|$.



Difference and Complement

- **Definition:** Let A and B be two sets. The **difference** of A and B, denoted by $A - B$ or $A \backslash B$, is the set containing those elements that are in A but not in B. It is also called the **complement of B with respect to A**
- $A - B = \{x \mid x \in A \wedge x \notin B\}$
- **Example:** $\{1, 3, 5\} - \{1, 2, 3\} = \{5\}$.
- **Definition:** Let U be the universal set. The **complement** of a set A, denoted by $\bar{A}$, is the complement of A with respect to U: $U - A$**.**
- $\bar{A} = \{x \mid x \in U \wedge x \notin A\}$
- When U is assumed, it is simplified to $\bar{A} = \{x \mid x \notin A\}$

Exercise on Difference and Intersection

- Which of the following is equivalent to the intersection $A \cap B$?

1. $((A \cup B) - A) - B$
2. $A - (A - B)$
3. $B - (A - B)$

- Correct answer is 2.
- Can you express the set difference $A - B$ using intersection and complement?
- $A - B = A \cap \bar{B}$

Set Identities

Set identities	
Identity	Name
$A \cup \emptyset = A$ $A \cap U = A$	Identity laws
$A \cup U = U$ $A \cap \emptyset = \emptyset$	Domination laws
$A \cup A = A$ $A \cap A = A$	Idempotent laws
$\overline{(\overline{A})} = A$	Complementation law
$A \cup B = B \cup A$ $A \cap B = B \cap A$	Commutative laws

Set Identities (cont'd)

Set identities	
Identity	Name
$A \cup (B \cup C) = (A \cup B) \cup C$ $A \cap (B \cap C) = (A \cap B) \cap C$	Associative laws
$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$	Distributive laws
$\overline{A \cup B} = \overline{A} \cap \overline{B}$ $\overline{A \cap B} = \overline{A} \cup \overline{B}$	De Morgan's laws
$A \cup (A \cap B) = A$ $A \cap (A \cup B) = A$	Absorption laws
$A \cup \overline{A} = U$ $A \cap \overline{A} = \emptyset$	Complement laws

Set Identities and Logic Equivalences

- All these set identities follow from corresponding logic equivalences
- Example: De Morgan's law

$$
\begin{array}{lcll}
\overline{A \cap B} & = & \{x | x \notin A \cap B\} & \text{by defn. of complement} \\
& = & \{x | \neg(x \in (A \cap B))\} & \text{by defn. of does not belong symbol} \\
& = & \{x | \neg(x \in A \wedge x \in B\} & \text{by defn. of intersection} \\
& = & \{x | \neg(x \in A) \vee \neg(x \in B)\} & \text{by 1st De Morgan law} \\
& & & \text{for Prop Logic} \\
& = & \{x | x \notin A \vee x \notin B\} & \text{by defn. of not belong symbol} \\
& = & \{x | x \in \overline{A} \vee x \in \overline{B}\} & \text{by defn. of complement} \\
& = & \{x | x \in \overline{A} \cup \overline{B}\} & \text{by defn. of union} \\
& = & \overline{A} \cup \overline{B} & \text{by meaning of notation}
\end{array}
$$

- So, just replace ∩ with ∧, ∪ with ∨, $\overline{\ }$ with ¬

Function

- **Definition:** Let A and B be nonempty sets. A **function** from A to B is an assignment of exactly one element of B to each element of A. We write $f(a) = b$, if b is the unique element of B assigned by the function f to the element a of A. If f is a function from A to B, we write $f: A \to B$.
- **Remark:** Functions are sometime also called **mappings** or **transformations**. If f is a function from A to B, we say that f **maps** A to B.
- **Definition:** If f is a function from A to B, we say that A is the **domain** of f and B is the **codomain** of f. If $f(a) = b$, we say that b is the **image** of a and a is the **preimage** of b. The **range** of f is the set of all images of elements of A.

Examples

- Let f be the function that assigns the last two bits of a bit string of length 2 or greater to that string. For example, $f(11010) = 10$. Then, the domain of f is the set of all bitstrings of length 2 or greater, and both the codomain and range are the set $\{00, 01, 10, 11\}$.
- Let $f: \mathbf{Z} \to \mathbf{Z}$ assign the square of an integer to this integer. Then $f(x) = x^2$, where the domain of f is the set of all integers.
- We take the codomain of f to be the set of all integers (or positive integers), and the range of f is the set of positive integers that are perfect squares, namely, $\{0, 1, 4, 9, \ldots\}$.

Injective Function

- **Definition**

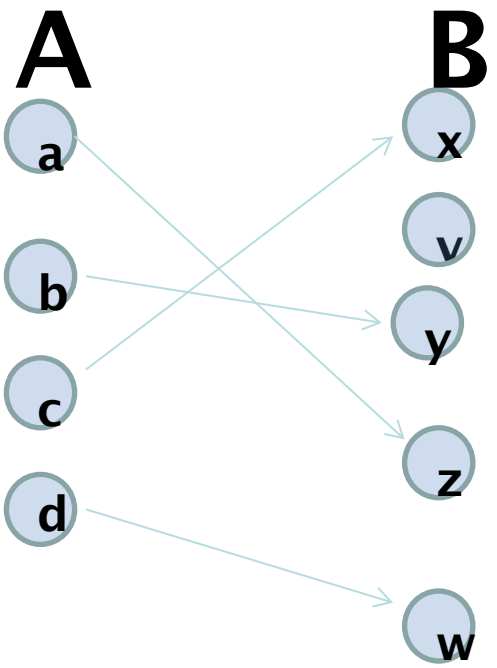
  A function f is said to be **injective** (or **one-to-one**) if and only if $f(a) = f(b)$ implies that $a = b$ for all a and b in the domain of f. An injective function is also called an **injection**.

- **Example**

  Is the function $f(x) = x + 1$ from the set of real numbers to the set of real numbers injective?

- **Example**

  Is the function $f(x) = x^2$ from the set of integers to the set of integers injective?



Exercise on Injective Functions

- Let f be a **one-to-one** (injective) function: $\forall a \forall b((f(a) = f(b)) \rightarrow (a = b))$. Which of the following statements is true for f.

1. $\forall a \forall b((a \neq b) \rightarrow (f(a) \neq f(b)))$

True because contrapositive of original implication

2. $\forall a \forall b((a = b) \rightarrow (f(a) = f(b)))$

True by the definition of functions

3. $\forall a \forall b((f(a) \neq f(b)) \rightarrow (a \neq b))$

True because contrapositive of 2.

Exercise on Functions

- Let *A* be a set with cardinality *n*
- Let *B* be a set with cardinality *n*
- Q1: What is the number of functions from A to B, assuming no constraints?
- Each element of *A* can be mapped to any of *n* elements of B; thus, *n* choices for first element, times *n* choices for second element,…,times *n* choices for *n*-th element. In total n^n functions.
- Q2: What is the number of *injective* functions from A to B?
- There are *n* choices for first element, times *n*-1 choices for second element,…,times 1 choice for *n*-th element. In total *n!* functions.

Surjective Function

- **Definition**

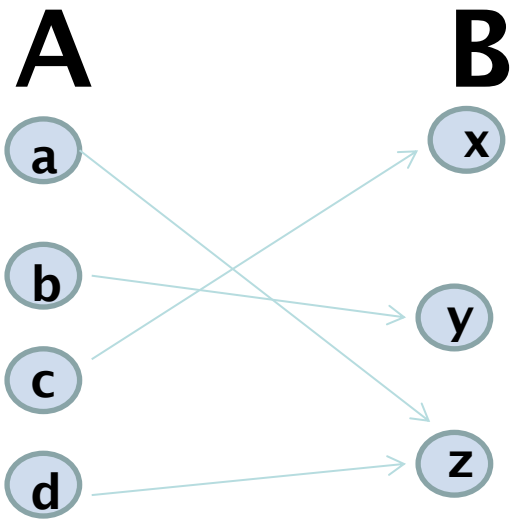
  A function f from set A to the set B is said to be **surjective** (or **onto**) if and only if for every element $b \in B$ there is an element $a \in A$ with $f(a) = b$. A surjective function is also called a **surjection**.

- **Example**

  Is the function $f(x) = x + 1$ from the set of integers to the set of integers surjective?

- **Example**

  Is the function $f(x) = x^2$ from the set of integers to the set of integers surjective?

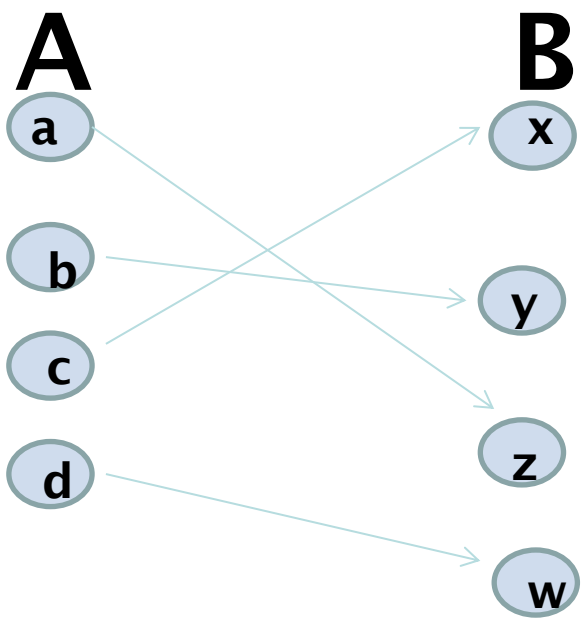


Bijection

- **Definition**

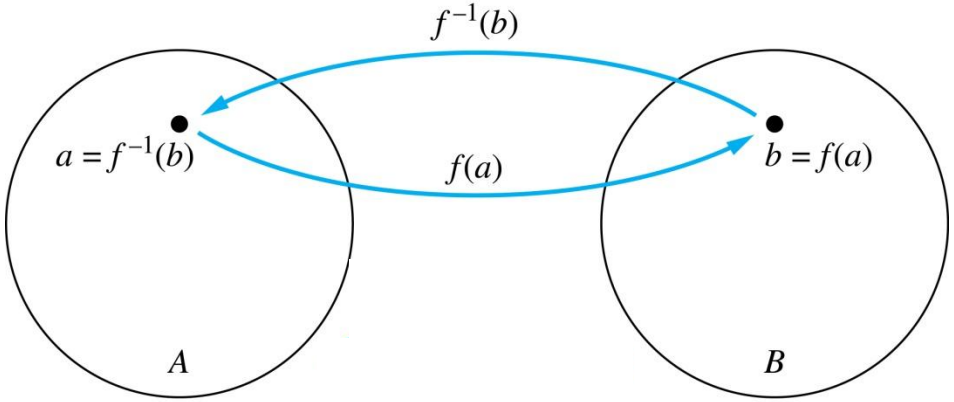
  A function f is a **bijection** (or *one-to-one correspondence*) iff it is **both one-to-one and onto**.

- Note:
  - one-to-one: injection
  - onto: surjection
  - one-to-one and onto: bijection



Inverse Function

- **Definition** Let f be a one-to-one correspondence (bijection) from the set A to the set B. The **inverse function** of f is the function that assigns to an element b belonging to B the unique element a in A such that $f(a) = b$. The inverse function of f is denoted by f^{-1}. Hence, $f^{-1}(b) = a$ when $f(a) = b$.
- A one-to-one correspondence is called **invertible**.



Examples

- **Example:** Let $f: \mathbf{Z} \to \mathbf{Z}$ be such that $f(x) = x + 1$. Is f invertible, and if it is, what is its inverse?

Yes. $f^{-1}(y)=y-1$

- **Example**

  Let $f: \mathbf{Z} \to \mathbf{Z}$ be such that $f(x) = x^2$. Is f invertible?

Not invertible because not one-to-one correspondence (e.g., $f(4)=f(-4) = 16$).

- **Example**

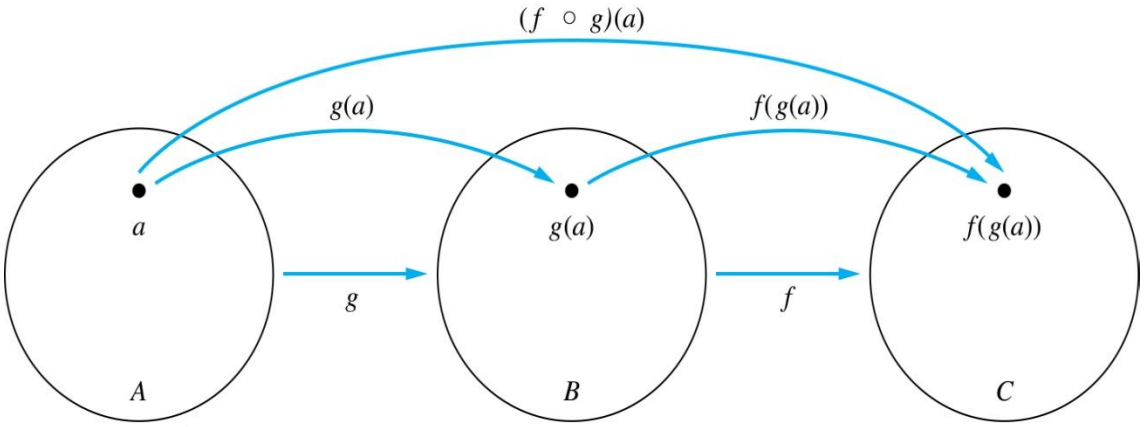
  Let $f: \mathbf{R}^+ \to \mathbf{R}^+$ be such that $f(x) = x^2$. Is f invertible?

Yes. $f^{-1}(y)=\sqrt{y}$

Composition

- **Definition**

  Let g be a function from the set A to the set B and let f be a function from the set B to the set C. The **composition** of the functions f and g, denoted by $f \circ g$, is defined by $(f \circ g)(a) = f\big(g(a)\big)$.



Composition

- **Example**
  Let f and g be functions from $\mathbf{Z}$ to $\mathbf{Z}$ defined by $f(x) = 2x + 3$ and $g(x) = 3x + 2$. What is the composition of f and g? What is the composition of g and f?
- $(f \circ g)(x) = f\big(g(x)\big) = f(3x + 2) = 2(3x + 2)+3 = 6x+7$
- $(g \circ f)(x) = g\big(f(x)\big) = f(2x + 3) = 3(2x + 3)+2 = 6x+11$
- Composition is not commutative

Cardinality of Infinite Sets

- **There are many infinities!**
- **Definition:** Two infinite sets A and B have the same cardinality if and only if there is a one-to-one correspondence from A to B.
- **Definition:** Cardinality $|\mathbf{N}| = \aleph_0$ of natural numbers $\mathbf{N}$ {0, 1, 2, …} is called *aleph-nought*
- A infinite set S with $|S| = \aleph_0$ is called is **countable.**
  - Set S is countable if and only if we can list its elements in sequence indexed by the natural numbers.
  - It is possible that $|S| = \aleph_0$ and $S \subset \mathbf{N}$ (e.g., set of non-negative even numbers $\subset \mathbf{N}$), or $\mathbf{N} \subset S$ (e.g., $\mathbf{N} \subset \mathbf{Z}$)
- An infinite set that is not countable is **uncountable.**
  - Which set has higher cardinality: $\mathbf{N}$ or the set of real numbers between 0 and 1?

Hilbert's Grand Hotel Paradox

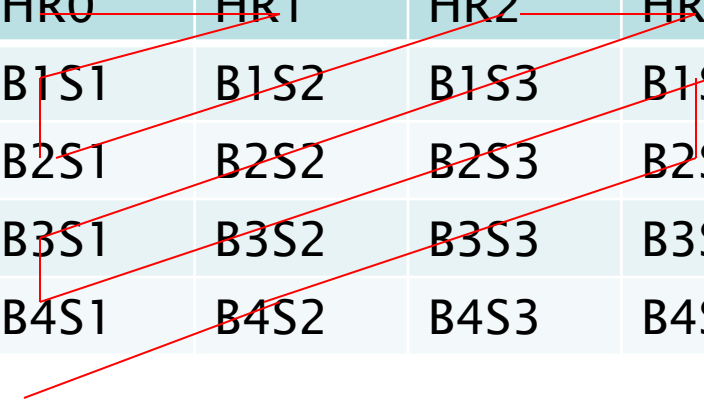
- Assume a hotel with infinite number of rooms, 0, 1,2,…, all full. Guest x is initially in room $r(x)$=x.
  - Although all rooms are full, we can still accept new quests
- A new guest arrives. How to accommodate her?
  - Move each existing guest to the next room $r(x) = x + 1$, so that the first room becomes available. This is possible because there are infinite rooms. Put new guest in Room 0.
- A bus with a finite number (e.g., 100) of new guests arrives. How to accommodate them?
  - Move each existing guest to room $r(x) = x + 100$, so that the first 100 rooms become available.
- A bus with an infinite but countable number of new guests arrives. How to accommodate them?
  - Move each existing guest to room $r(x) = 2x$, so that all odd numbered rooms (countable) become available.

https://www.youtube.com/watch?v=OxGsU8oIWjY&ab_channel=Veritasium

Hilbert's Grand Hotel (cont)

- Assume a hotel with infinite number of rooms, 0, 1,2,… all full. Guest x is initially in room $r(x)=x$.
- An infinite countable number of buses, each with an infinite countable number of new guests arrives. How to accommodate them?

Hotel	HR0	HR1	HR2	HR3
Bus 1	B1S1	B1S2	B1S3	B1S4
Bus 2	B2S1	B2S2	B2S3	B2S4
Bus 3	B3S1	B3S2	B3S3	B3S4
Bus 4	B4S1	B4S2	B4S3	B4S4



- Create a room sequence by following the red line: HR0, HR1, B1S1, B2S1, B1S2, HR2, HR3, …

Hilbert's Grand Hotel (cont)

- Assume a hotel with infinite number of rooms, 0,1,2,… all full. Guest x is initially in room $r(x)=x$.
- An (uncountable) number of guests arrives. Each guest has a unique ID which is a string of 0,1. How to accommodate them?
- Assume that you can accommodate all guests in increasing order of ID.

Room	Guest
Room 1	000011111..
Room 2	010001111..
Room 3	011001010
…	…

- I generate a guest ID 100…, by flipping the first bit of the first guest, the second bit of the second guest etc.
  - This new ID 100… is different from every other on (at least) the diagonal, i.e., different from ID1 on first bit, from ID2 on second bit etc.
  - No matter what I do, I can always find a guest that cannot be accommodated.

Example of Countable Sets

- To show that an infinite set S is countable, we can find a one-to-one correspondence between $\mathbf{N}$ and S.
- $\mathbf{Z}^+ = \{1, 2, 3, \ldots\}$.
  Let $f: \mathbf{N} \to \mathbf{Z}^+$ be $f(x) = x + 1$

N	0	1	2	3	..
	|	|	|	|	
Z^+	1	2	3	4	..

- The set of nonnegative even numbers.

$$f(x) = 2x$$

N	0	1	2	3	..
	|	|	|	|	
even Z^+	0	2	4	6	..

- The set of all integers $\mathbf{Z}$.
- List all integers as: $0, 1, -1, 2, -2, 3, -3, \ldots$

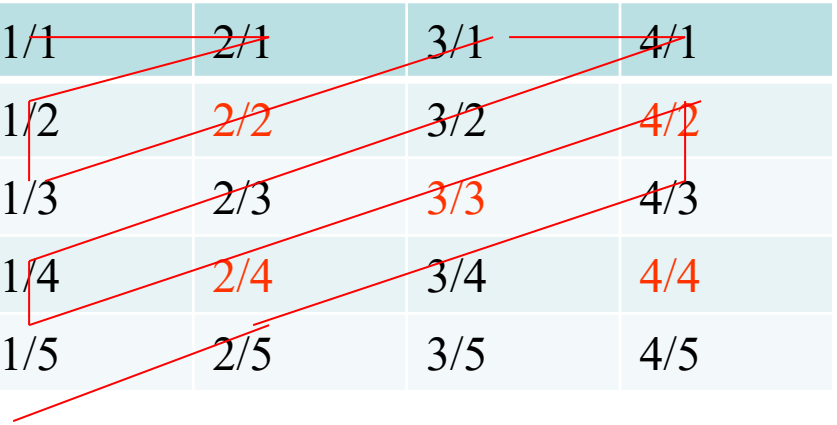
Define $f(x) = \begin{cases} -\frac{x}{2}, if\ x\ is\ even \\ \frac{x+1}{2}, if\ x\ is\ odd \end{cases}$

N	0	1	2	3	4	..
	|	|	|	|		
Z	0	1	-1	2	-2	..

Set of Positive Rational Numbers

- Every positive rational number can be written as the quotient p/q of two positive integers
- Arrange rational numbers in 2D array, where the first row contains numbers with q=1, the second row q=2, etc. Within the same row, arrange in increasing order of p.

1/1	2/1	3/1	4/1
1/2	2/2	3/2	4/2
1/3	2/3	3/3	4/3
1/4	2/4	3/4	4/4
1/5	2/5	3/5	4/5



- Red line lists rational numbers in increasing order of $p+q$: 1/1, 2/1, 1/2, 1/3, 2/2, 3/1 etc. Do not list duplicates.
- Is the set of *all* rational numbers countable?
  - Yes: If A and B are countable, then $A \cup B$ is countable

Set of Real Numbers Between 0 and 1

- Assume that this set is countable. Then, the set of all real numbers between 0 and 1 can be listed in some order

r_1=0.$d_{11}d_{12}d_{13}$ ..

r_2=0.$d_{21}d_{22}d_{23}$ ..

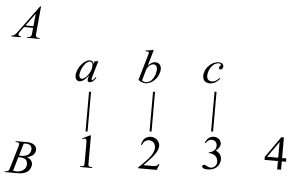
r_3=0.$d_{31}d_{32}d_{33}$ ..

…, where d_{ij} is a digit between 0 and 9.

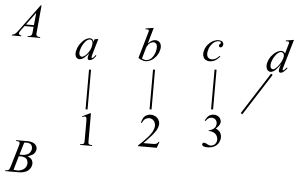
- I generate a real number r=0.$d_1d_2d_3$.. such that $d_1 \neq d_{11}, d_2 \neq d_{22}, d_3 \neq d_{33}$..
- r is different from each r_i on digit d_i, contradicting the assumption that the set of all real numbers between 0 and 1 can be listed in some order.
- Thus, this set is uncountable.
- This type of proof is called Cantor diagonalization argument

Recap on Functions

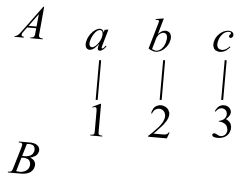
A function $f:A \to B$ is **injection** (or **one-to-one**) if and only if $f(a) = f(b)$ implies that $a = b$ for all a and b in the domain of f.



A function $f:A \to B$ is **surjection** (or **onto**) if and only if for every element $b \in B$ there is an element $a \in A$ with $f(a) = b$.



A function $f:A \to B$ is a **bijection** (or *one-to-one correspondence*) iff it is **both one-to-one and onto**.



Recap on Countable Sets

- To prove that S is **countable** we find a bijection f:$\mathbf{N} \to S$, or show how its elements can be listed in sequence indexed by the natural numbers
- Example: Set E of nonnegative even numbers is countable.
- Define function $f(a) = 2a$, $\mathbf{N} \to E$ and show that it is bijection.
- Assume $f(a) = f(b)$. Then, $2a = 2b \Rightarrow a = b$. Thus, f injection.
- Assume any number $a' \in E$. Since a' is even: $\exists k \in N\ a' = 2k = f(k)$. Thus, f surjection.
- Thus, f bijection.
- Countable Sets: Integers, Rational Numbers
- Uncountable Sets: Real Numbers
- The continuum hypothesis: There is no set whose cardinality is strictly between that of natural numbers and that of real numbers (Hilbert's first problem)

Properties of Functions and Cardinality of Infinite Sets

For two sets A and B

- If there is bijection $f: A \to B$, then $|A| = |B|$ (**Bijection principle**)
- If there is injection $f: A \to B$, then $|A| \leq |B|$
- If there is injection $f: A \to B$, but not bijection, then $|A| < |B|$

Schröder-Bernstein Theorem: if $|A| \leq |B|$ and $|B| \leq |A|$, then $|A| = |B|$.

- Example: Show that $|(0,1)| = |(0,1]|$
  - First interval contains all real numbers x: $0<x<1$
  - Second interval contains all real numbers x: $0<x\leq 1$
  - Not easy to find a bijection between the two sets. Easier to find two injections and apply the theorem.
- An injection $(0,1) \to (0,1]$: $f(x) = x$. Thus, $|(0,1)| \leq |(0,1]|$
- An injection $(0,1] \to (0,1)$: $f(x) = x/2$. Thus, $|(0,1]| \leq |(0,1)|$
- Thus, according to theorem: $|(0,1)| = |(0,1]|$

Additional Theorems

- Theorem. The cardinality of any set S is lower than the cardinality of its powerset $P(S)$: $|S| < |P(S)|$
  - It holds for both finite and infinite sets
  - $P(\mathbf{R})$ has higher cardinality than the set of real numbers $\mathbf{R}$
- Definition: A function is computable if there is a computer program that finds the value of the function on any input. Otherwise, it is uncomputable.
- Theorem: There exist functions from $\mathbf{N}$ to {0, 1} that are uncomputable.
  - Each function $\mathbf{N}$ to {0, 1} can be mapped to an infinite bitstring
    - $f_1(0)=0$, $f_1(1)=0$, $f_1(2)=1$, $f_1(3)=0$.. maps to 0010..
    - $f_2(0)=0$, $f_1(2)=1$, $f_1(2)=0$, $f_1(3)=1$.. maps to 0101..
    - The set of infinite bitstrings (and functions) is uncountable
  - Each program can be encoded as a finite bitstring. The set of all finite bitstrings is countable, Thus, the set of all possible computer programs is also countable.
- Halting Problem (Alan Turing): There is no general program that determines whether another program terminates for some input